

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behaviour. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify its main effect, and identifies its interaction with credit access only off the eight adopter economies in that sample; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access

is robustly associated with green practice adoption (average marginal effect of 13 percentage points in the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO₂-emissions-monitoring outcome and with country fixed effects the primary sample’s single-wave design cannot support informatively, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest’s feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a nominally significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, though this does not survive a Benjamini–Hochberg correction for multiple comparisons across the full family of interaction tests reported in the paper, and we treat it as suggestive rather than established. We interpret the combined evidence as more consistent with green banking policy’s binding constraint operating through

firm-level absorptive capacity than through the country-level regulatory capacity its own institutional design targets, though we report this as our best current account of an auxiliary, exploratory finding rather than as a settled result. Finally, because none of the above designs exploits exogenous variation in SBFN adoption *timing*, we additionally build a country-year panel (217 economies, 2000–2024) from World Bank World Development Indicators and estimate a Callaway-Sant’Anna staggered event-study around each economy’s own adoption date – the identification strategy this literature most often calls for. It shows no effect on renewable-energy share and a positive association with CO2 emissions per capita that shrinks by roughly 45% once GDP per capita is controlled for and whose event-study profile is more consistent with pre-existing differences in adopting economies’ growth trajectories than with a causal effect of adoption; this independent, differently-identified test converges on the same substantive conclusion as the firm-level analysis.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner pro-

7 duction. On paper, this represents one of the most ambitious diffusions of
8 environmental financial regulation outside the OECD: economies as differ-
9 ent in income, banking-sector depth, and political regime as Bangladesh,
10 Morocco, Mongolia, and Indonesia now report, to the International Finance
11 Corporation’s Sustainable Banking and Finance Network (SBFN), formal
12 sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-
13 risk disclosure rules for lenders. Whether that adoption changes what actu-
14 ally happens on a factory floor—whether a firm that can get a bank loan is
15 any more likely to install a wastewater treatment system, switch to a more
16 efficient boiler, or measure its own emissions—is a separate and considerably
17 harder question, and it is the one this paper answers.

18 The premise that formal adoption of a policy is not the same as its op-
19 eration in practice is, of course, not new to institutional economics. A large
20 literature on state capacity and policy credibility has long argued that the
21 same law, regulation, or guideline can produce entirely different real-economy
22 outcomes depending on the administrative and judicial infrastructure avail-
23 able to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to
24 green finance specifically, a parallel body of work has documented that green
25 credit guidelines in China measurably reshape lending allocation and firm
26 environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al.,
27 2025), that firms’ green investment responds to the financing constraints they
28 face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and
29 that the World Bank Enterprise Surveys’ own Green Economy module is a
30 workable lens on firm-level environmental behaviour across dozens of coun-
31 tries at once (Asif et al., 2025; Phan, 2026). What is missing is the connection

32 between these two literatures: a direct test, at firm level and across a wide
33 cross-section of the very economies the green-banking-policy movement tar-
34 gets, of whether the credit-to-green-investment channel that these guidelines
35 are designed to activate is actually stronger where the guidelines exist—and
36 whether that depends on the state capacity to make the guideline credible in
37 the first place.

38 This paper closes that gap using firm-level data from two independently
39 assembled samples of World Bank Enterprise Survey (WBES) waves. The
40 primary sample covers 47 country-years fielded between 2018 and 2023 that
41 carry the Bank’s full Green Economy module—a battery of questions on emis-
42 sions monitoring, energy audits, and the adoption of climate-friendly produc-
43 tion measures—alongside the Surveys’ standard access-to-finance module; a
44 second, global sample extends this to 162 further country-years fielded be-
45 tween 2021 and 2026, reaching for the first time in this literature into high-
46 income economies with no SBFN involvement at all (Section 4). We match
47 each economy to its Sustainable Banking and Finance Network policy status
48 as of its survey year, distinguishing genuine ex-ante policy adopters from
49 economies that joined only afterward, and to the World Bank’s Worldwide
50 Governance Indicators as a measure of the regulatory state capacity avail-
51 able to make that policy credible. Econometrically, we depart deliberately
52 from the pooled fixed-effects specification that has become the default in
53 this literature. Because SBFN status is fixed within country in a single-
54 wave cross-section, a country fixed effect absorbs its main effect entirely, and
55 would identify its interaction with credit access—which remains estimable,
56 being a firm-level term—only for the eight adopter economies in that sample;

57 a naive interaction term estimated by pooled logit, which we report first as
 58 a baseline precisely to make this limitation visible rather than to hide it,
 59 returns no detectable policy-moderation effect at all. We show that this null
 60 result is not evidence of an absent relationship so much as a symptom of
 61 forcing a fundamentally hierarchical question (does a level-2, country-level
 62 policy variable reshape a level-1, firm-level financing relationship?) into a
 63 flat, single-level model with too few country clusters to lean on for inference
 64 (Cameron and Miller, 2015). We then re-estimate the same relationship with
 65 tools built for exactly this structure: a Bayesian hierarchical logistic model
 66 with country-varying slopes (Gelman and Hill, 2007; Gelman et al., 2020), a
 67 causal-forest / double-machine-learning (DML) estimator (Wager and Athey,
 68 2018; Chernozhukov et al., 2018; Athey et al., 2019; Athey and Imbens, 2019)
 69 that lets the treatment effect of credit access vary flexibly across the institu-
 70 tional landscape rather than through a single estimated coefficient, and, on
 71 the much larger global sample, a country-fixed-effects specification that the
 72 primary sample’s single wave cannot support with useful precision.

73 This paper contributes to the literature in six respects. First, it pro-
 74 vides what is, to our knowledge, the first direct empirical test of whether
 75 SBFN-style green banking policy adoption changes the firm-level relation-
 76 ship between finance and green investment, rather than only the aggregate
 77 volume of green lending reported by banks themselves—a firm-level, demand-
 78 side complement to the bank-level, supply-side evidence that dominates the
 79 existing green-credit-policy literature. Second, it extends the World Bank
 80 Enterprise Survey’s Green Economy module—used until now mostly within
 81 single countries or single regional waves—into a harmonized 41-country pri-

82 mary sample spanning Europe, Central Asia, the Caucasus, and the Middle
 83 East and North Africa, giving the paper a breadth of institutional varia-
 84 tion that a single-country or single-region design cannot offer. Third, and
 85 unusually for this literature, we subject the primary sample’s central find-
 86 ing to an independently assembled second test: a 162-economy global sam-
 87 ple built from a World Bank standardized cross-country database spanning
 88 2021–2026 that, to our knowledge, no published study of green banking pol-
 89 icy has yet exploited, extending coverage for the first time into high-income
 90 economies entirely outside the SBFN’s own remit and allowing a country-
 91 fixed-effects specification the primary sample’s single-wave design cannot
 92 support. That the same qualitative result—a robust finance-green channel,
 93 no detectable institutional moderation—survives this independent replica-
 94 tion is, we think, considerably more informative than either sample could
 95 be alone. Fourth, methodologically, the paper demonstrates—with the same
 96 underlying question tested twice, on two different samples, under four differ-
 97 ent estimators—exactly where and why a classical fixed-effects approach fails
 98 to detect (or, when country fixed effects are usable, correctly fails to find) an
 99 institutionally-conditioned effect, offering applied researchers working with
 100 similarly structured cross-country firm data a concrete illustration of when
 101 the extra modelling investment is worth making and when a null result is sim-
 102 ply a null result. Fifth, the paper redirects the question of which system-level
 103 margin governs green-credit-policy effectiveness: having ruled out a country-
 104 institutional interaction across four estimators and two samples, we show
 105 (Section 3.4) that the effect heterogeneity that does exist is concentrated at
 106 the level of firm size rather than country institutions, relocating the opera-

tive “system condition” in the Policy Design $\times$ System Conditions $\rightarrow$ System
 Outcomes logic this special issue organizes itself around from the country-
 regulatory level to the firm-scale level—a mechanism we introduce as an
 auxiliary, exploratory hypothesis in Section 3.4 and demonstrate empirically
 in Section 5.4 rather than treat as a pre-registered finding, and are corre-
 spondingly explicit about the identification limits (Section 4) that a purely
 cross-sectional design, however extensively replicated, still carries. Sixth, and
 directly in response to that identification-strategy limit, Section 6 builds an
 entirely independent country-year panel from World Bank World Develop-
 ment Indicators and estimates a Callaway-Sant’Anna staggered event-study
 exploiting SBFN’s own adoption-timing variation—the design class this spe-
 cial issue’s methodological guidance names first—and finds that it, too, does
 not support a beneficial institutional effect on aggregate environmental out-
 comes, converging with the firm-level evidence from a design built specifically
 to meet an identification bar the firm-level data cannot.

The paper also speaks directly to a growing body of recent scholarship in
 this journal on the institutional and financial determinants of firm and bank
 behaviour in emerging and transition economies. Yan et al. (2024) study
 how environmental protection taxes reshape green productivity among Chi-
 nese listed firms; Barra and Falcone (2026) examine how institutional factors
 condition environmental performance across a global sample of economies;
 Heyert and Weill (2025) and Bach et al. (2025) use firm- and bank-level data
 across dozens of countries to study, respectively, trust-driven financial inclu-
 sion and credit constraints’ effect on corporate tax behaviour in Vietnamese
 firms; Li et al. (2026) study how shadow-banking engagement reshapes trade-

132 credit financing among Chinese non-financial firms; and Horvath et al. (2025)
133 apply Bayesian model averaging to the cross-country determinants of finan-
134 cial development, a methodological precedent we build on directly in adopting
135 a Bayesian estimation strategy in Section 4. This paper sits at the intersec-
136 tion of these threads—institutions, green performance, and firm-level financ-
137 ing behaviour, studied jointly rather than pairwise—and, to our knowledge,
138 is the first among them to test whether a specific, dateable policy-adoption
139 event (SBFN membership) moderates the finance-green relationship at firm
140 level across a comparably broad cross-section of adopting economies.

141 The remainder of the paper is organized as follows. Section 2 sets out
142 the institutional background of the SBFN framework and the concepts and
143 trends in green banking policy adoption across our sample. Section 3 devel-
144 ops the paper’s three hypotheses from the state-capacity and institutional-
145 complementarity literature. Section 4 describes the primary and global sam-
146 ples, variable construction, and the empirical strategy for each. Section 5
147 presents baseline, hierarchical, and heterogeneous-effect results on the pri-
148 mary sample, followed by the independent global-sample replication. Sec-
149 tion 7 discusses the findings’ economic magnitude and their implications for
150 how green credit policy is designed and sequenced, and Section 8 concludes.

151 **2. Institutional background: green banking policy in emerging and** 152 **transition economies**

153 The Sustainable Banking and Finance Network (SBFN)—established in
154 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
155 expanded mandate covering both banking and non-bank financial regulation—

156 is the principal vehicle through which financial-sector regulators and banking
157 associations in emerging and developing economies have coordinated the de-
158 sign of national sustainable-finance policy. Hosted and technically supported
159 by the International Finance Corporation, the network now counts roughly
160 seventy member institutions (central banks, banking regulators, and banking
161 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
162 East Asia and the Pacific, Europe and Central Asia, Latin America and
163 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
164 national Finance Corporation / Sustainable Banking and Finance Network,
165 2025). Notably for our sample, the network’s membership is essentially dis-
166 joint from the European Union: not one of the sixteen EU member states in
167 our WBES sample appears on any SBFN membership roster we could locate,
168 consistent with the network’s explicit mandate to serve markets outside the
169 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
170 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
171 most of our survey waves in any case).

172 Membership itself, however, is only the network’s coarsest measure of
173 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
174 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
175 2023–2024) and country-level Progress Report addenda—scores each mem-
176 ber jurisdiction’s actual regulatory output against several policy pillars, most
177 centrally environmental-and-social risk management and, in later report vin-
178 tages, ESG disclosure integration and green-finance-flow measurement, and
179 places each jurisdiction on a progression from an initial “Commitment” or
180 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-

ing” and, for the most mature frameworks, “Consolidating.” The variation
 this produces across our sample is considerable and is itself part of the paper’s
 identifying variation, not just a robustness afterthought. Mongolia’s Bank
 of Mongolia was a founding SBFN member in 2012 and had already reached
 the network’s “Advancing” implementation stage by the time of the country’s
 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
 member, had progressed by its 2022 Enterprise Survey wave into the net-
 work’s ESG-integration pillar at an “Advancing” classification, while India’s
 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
 had progressed only as far as an initial “Commitment” stage by their 2019
 survey waves. A further, and for identification purposes equally important,
 group of economies in our sample joined SBFN only after their Enterprise
 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
 istan (2023), and most of the Western Balkan and South Caucasus economies
 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
 their survey year, even though several are now members. This ex-ante cod-
 ing is deliberate: it is the policy environment a firm actually operated under
 at the moment its Green Economy module responses were recorded that is
 relevant to our test, not the policy environment that jurisdiction eventually
 adopted.

The economic logic behind the SBFN framework, and the reason it is a
 plausible channel for firm-level green investment rather than a purely sym-

206 bolic commitment, rests on a straightforward supervisory mechanism: once
 207 a banking regulator formally adopts environmental-and-social risk manage-
 208 ment guidelines, commercial banks under its supervision face a compliance
 209 incentive to price environmental risk into individual lending decisions and,
 210 in many of the more advanced frameworks, to report a green-lending share
 211 to the supervisor directly. Existing evidence on China’s much-studied green
 212 credit guidelines documents exactly this channel operating at bank and firm
 213 level—reduced lending to high-polluting borrowers and improved environ-
 214 mental performance among firms that retain access to credit (Li et al., 2023;
 215 Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller liter-
 216 ature on Bangladesh’s own long-running green-banking guidelines points in
 217 the same direction (Khairunnessa et al., 2021). What neither literature has
 218 established is whether this channel is detectable across a broad, heteroge-
 219 neous cross-section of SBFN-style adopters at the firm level using a common
 220 survey instrument, nor whether the strength of the channel depends on the
 221 regulatory capacity available to enforce the underlying guideline—the two
 222 questions this paper turns to next.

223 **3. Literature review and hypothesis development**

224 *3.1. Access to finance and firm-level green investment*

225 A firm’s decision to install pollution-abatement equipment, retrofit ma-
 226 chinery for energy efficiency, or absorb the upfront cost of an emissions-
 227 monitoring system is, first and foremost, an investment decision, and in-
 228 vestment decisions are shaped by whether a firm can finance them. The
 229 theoretical case is close to a standard credit-constraint story: green invest-

230 ments of the kind captured in the WBES Green Economy module (heat-
 231 ing and cooling upgrades, on-site renewable generation, waste-minimization
 232 retrofits, energy-management systems) typically carry a large upfront capi-
 233 tal outlay against a payback stream realized only over several years, which is
 234 exactly the profile of investment most sensitive to a firm’s access to external
 235 finance rather than to retained earnings alone. Firm-level evidence bears
 236 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
 237 facturing firms, show that credit-constrained firms are measurably less likely
 238 to adopt cleaner production practices even when they report an equivalent
 239 willingness to comply with environmental regulation. De Haas et al. (2025)
 240 document, across a large cross-country firm sample, that managerial and fi-
 241 nancial barriers operate as distinct but jointly binding constraints on firms’
 242 green transition investments, and Kalantzis et al. (2022) reach a closely re-
 243 lated conclusion using EBRD transition-economy survey data, finding that
 244 firms’ green investment is driven by a mix of financing conditions and cli-
 245 mate exposure rather than by climate motivation alone—precisely the joint
 246 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
 247 and Truong (2026) find, respectively in OECD and Vietnamese data, that
 248 financing constraints blunt firms’ green investment response to environmen-
 249 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 250 (2023) show that firms with better access to credit are both more likely to
 251 have adopted green management practices before a shock and better able
 252 to sustain them through one. Ullah (2025) narrows this further to unlisted
 253 SMEs specifically, showing that financial constraints bind corporate environ-
 254 mental responsibility at smaller firm scales—a firm-size channel this paper’s

own causal-forest evidence (Section 5) also implicates, though, as we report there, without recovering a clean monotonic size gradient of its own. Siewers et al. (2024) extend the firm-level environmental-performance literature to a global-value-chain setting, showing that a firm’s position in international production networks shapes its environmental performance net of domestic financing conditions, and Bambe et al. (2026) show that environmental policy stringency itself reshapes firm efficiency in developing economies, underscoring that the firm-level margin, not only the aggregate or bank-level one, is where much of this literature’s true test now lies. Taken together, this literature converges on a simple first-order proposition, which any test of a green-credit-policy channel must first establish before asking whether policy moderates it:

H1. Firms with access to formal bank credit (a credit line or loan from a financial institution) are more likely to have adopted green production practices than otherwise-similar firms without such access.

3.2. *Does green banking policy strengthen the finance-to-green-investment channel?*

H1 describes a channel, not a policy. The entire premise of the SBFN framework and its national analogues—and of the much larger literature on green credit guidelines in China specifically (Li et al., 2023, 2024; Huang et al., 2023)—is that a supervisory mandate directing banks to price environmental risk into lending, or to preferentially finance green activity, should widen the gap in green-investment behaviour between firms with and without bank access relative to a counterfactual world without the policy: banks

280 in adopting jurisdictions have a compliance incentive to seek out, and price
 281 favourably, exactly the borrowers undertaking the kind of investment our
 282 outcome variable measures, while credit-constrained firms in the same ju-
 283 risdiction see no comparable improvement in their financing terms. This
 284 is a moderation hypothesis, not a simple main-effect one, and it is worth
 285 being explicit that most of the existing supporting evidence—again, over-
 286 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
 287 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
 288 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
 289 is drawn from single-country studies of a single, unusually well-resourced
 290 green-credit regime. Whether the same interaction holds across the far more
 291 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
 292 work, many of them at a far earlier and less resourced stage of implementation
 293 than China’s, is untested. This motivates:

294 **H2.** The positive association between bank credit access and
 295 green-practice adoption (H1) is stronger in economies that had,
 296 as of the survey year, adopted an SBFN sustainable-finance policy
 297 framework than in economies that had not.

298 *3.3. Institutional complementarity: does policy credibility depend on state*
 299 *capacity?*

300 A green banking guideline is only as effective as the supervisory apparatus
 301 behind it. This is the institutional-complementarity argument at the centre of
 302 the comparative-institutions, law-and-finance, and state-capacity literatures.
 303 Hall and Soskice (2001)’s varieties-of-capitalism framework is the founda-
 304 tional statement of the underlying logic in its most general form: formally

305 similar policy instruments produce systematically different outcomes depend-
 306 ing on how they interlock with the surrounding institutional architecture—
 307 a regulatory mandate that functions as intended within one configuration
 308 of complementary institutions may function quite differently, or not at all,
 309 transplanted into another. Applied more narrowly to financial regulation
 310 specifically, the same logic runs through the law-and-finance and state-capacity
 311 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
 312 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
 313 2005, 2008) established that formally identical financial-contracting rules pro-
 314 duce very different real-economy financing outcomes depending on the quality
 315 of the legal and regulatory institutions that enforce them, while Acemoglu
 316 et al. (2001) showed the same logic operating at the level of a country’s en-
 317 tire institutional endowment. Applied to our setting, the argument is direct:
 318 a bank supervisor’s environmental-risk guideline can only reshape lending
 319 behaviour to the extent that the supervisor has the regulatory capacity—
 320 examination authority, credible sanctions, reliable disclosure enforcement—
 321 to make compliance with it more than nominal. Where that capacity is
 322 weak, an SBFN framework may exist on paper (a formal “Commitment” or
 323 “Formulating” stage classification, in the network’s own terminology) with-
 324 out meaningfully altering bank lending behaviour on the ground, precisely
 325 the state-capacity/policy-credibility distinction that motivates this special
 326 issue. This is also the logic behind the small but growing strand of work on
 327 institutions and environmental performance published in this journal specifi-
 328 cally: Barra and Falcone (2026), studying a global sample of economies, find
 329 that institutional-quality indicators condition environmental performance in

ways that are not reducible to income level alone, a result our design is built to test at the firm rather than the country level. We proxy regulatory capacity with the World Bank’s Worldwide Governance Indicators Regulatory Quality estimate, matched to each economy’s survey year, and hypothesize a three-way interaction:

H3. The policy-moderation effect described in H2 is itself increasing in regulatory quality: SBFN adoption strengthens the credit-to-green-investment channel most where the supervising jurisdiction has the institutional capacity to make the underlying guideline credible, and least—potentially not at all—where it does not.

Hypotheses H2 and H3 are, by construction, statements about how a country-level variable reshapes a firm-level relationship. That two-level structure is what Section 4 argues a classical single-level specification is poorly placed to adjudicate on this sample—not because the interaction terms it tests are unidentified, but because a country-cluster count in the low forties makes cluster-robust inference on them unreliable, and because a country-fixed-effects variant would purchase within-country identification at the cost of the level-2 main effects and of resting on only eight adopter economies. That is what motivates the hierarchical and machine-learning estimators we turn to after reporting the classical benchmark.

351 3.4. *An auxiliary hypothesis: firm-level absorptive capacity as an alternative*
352 *margin*

353 H2 and H3 are both, in the end, statements that a *country-level* char-
354 acteristic conditions the strength of the credit-to-green channel. A separate
355 strand of the financing-constraints literature suggests the conditioning vari-
356 able that matters most may instead sit at the *firm* level: Ullah (2025) shows
357 that financial constraints bind corporate environmental responsibility specif-
358 ically at smaller firm scales, consistent with a standard fixed-cost logic in
359 which the up-front capital outlay a green retrofit requires (Section 3) is pro-
360 portionately heavier, and therefore harder to absorb even once financed, for
361 a smaller firm than a larger one. We did not set out to test this as a fourth
362 ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s
363 feature-importance decomposition (Section 5.4), which we report honestly
364 here as an auxiliary, exploratory hypothesis motivated by that same fixed-
365 cost logic rather than one we pre-specified:

366 **H4 (auxiliary, exploratory).** Whatever heterogeneity exists
367 in the strength of the credit-to-green-adoption channel is more
368 attributable to firm size than to either country-level institutional
369 moderator (SBFN status or regulatory quality).

370 We flag its exploratory status explicitly because it changes how its evi-
371 dence should be read: Section 5.4’s support for H4 is genuine and, we think,
372 informative, but it is evidence discovered during analysis rather than con-
373 firmed against a pre-registered prediction, and the same section reports hon-
374 estly that it does not yet extend to a clean, monotonic size gradient. H4

also connects this paper’s evidence to the just-transition and SME-financing literature’s broader concern with who is left behind when a policy instrument operates on an extensive banking margin rather than on the firm-scale margin that determines whether a financed firm can absorb a green investment’s fixed costs at all—a distributional dimension of green-credit-policy design this paper’s data can only gesture toward, not test directly, but which we think is a natural extension for future work using firm-level data with a continuous size measure and a larger, more granular firm-size distribution than the Enterprise Surveys’ own three-category strata provide.

4. Data and methodology

4.1. Data sources and sample construction

The paper’s firm-level data come from the World Bank Enterprise Surveys (WBES), a standardized, stratified-random-sample survey of formal-sector, non-agricultural, non-extractive private firms with at least five employees, conducted by the World Bank’s Enterprise Analysis Unit. Our primary analysis sample is the World Bank’s own harmonized cross-country file combining the 41 Europe and Central Asia (ECA) and Middle East and North Africa (MENA) economies surveyed under a single, identical questionnaire between 2018 and 2020, the survey wave in which the Bank’s Green Economy module—a battery of questions on emissions monitoring, energy-consumption targets, and the adoption of specific climate-friendly production measures—was fielded in full to every respondent. This yields 28,042 firm-level observations across 41 economies (Table 2). We supplement this primary sample with 15,556 additional firm observations from six further economies

399 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
400 Timor-Leste 2021) in which a shortened, partially randomized version of the
401 Green Economy module was fielded; because the exact item wording and ran-
402 domized sub-module assignment differ across these six economies, we do not
403 pool them into the main harmonized outcome variable, and instead use them
404 as an out-of-region external-validity check on a narrower, honestly-matched
405 pair of indicators (CO2-emissions monitoring, available in five of the six;
406 waste-minimization adoption, available in five of the six, with different spe-
407 cific economies covered by each), reported separately in Section 5.

408 The primary sample, while broad by the standards of the existing green-
409 credit-policy literature, is not the full universe of WBES economies carrying
410 *any* Green Economy content, and we build a second, independent sample
411 to exploit the rest of it. The World Bank also maintains a standardized
412 cross-country database (2006–2026) that harmonizes a common set of core
413 variables across every Formal Sector Enterprise Survey wave it has ever
414 fielded, 292,314 firm-level observations across 471 country-years. Within
415 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
416 into high-income economies entirely outside the SBFN’s remit such as Aus-
417 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
418 Economy item set: climate-damage exposure and, more relevantly for our
419 outcome construct, CO2-emissions monitoring, alongside the same standard
420 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
421 stacle) used in the primary sample. We supplement this with Bangladesh
422 2022 and Indonesia 2023 — two economies whose Green Economy responses
423 used an alternative randomized-module variable naming convention and so

424 are absent from the standardized database’s harmonized columns, but which
425 we can recover from our own extraction of the underlying country files —
426 for a combined global sample of 162 economies and 100,115 firms (89,797
427 with complete data on all covariates). Firm size, sector, and productivity
428 controls for this sample are drawn from the Bank’s companion Productivity
429 Database, merged by firm identifier. We refer to this as the *global sample*,
430 and to CO2-emissions monitoring as its outcome variable throughout; it is
431 the paper’s vehicle for testing whether the primary sample’s central find-
432 ing replicates independently, at far greater scale, and under an identification
433 strategy — country fixed effects — that the primary sample’s single-wave
434 design cannot support (Section 4).

435 Country-level data are drawn from two sources for both samples. For the
436 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
437 and progression-framework stage as of each economy’s survey year are com-
438 piled from the Sustainable Banking and Finance Network’s Global Progress
439 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
440 denda, and the network’s live country-profile data portal, cross-checked against
441 IFC press releases announcing individual member accessions; economies that
442 joined SBFN only after their WBES survey year are coded as non-adopters
443 as of that year (Section 2). For the global sample’s remaining 157 economy-
444 years, we draw membership status and join year directly from the SBFN data
445 portal’s full published member roster (67 members as of the portal’s most
446 recent update), applying the identical ex-ante rule: an economy is coded as
447 an adopter only if its join year precedes its survey year. Six economy-years
448 in the global sample (Central African Republic, Cameroon, Chad, Republic

449 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
 450 the roster under a regional entry, “Central African States,” which we inter-
 451 pret as referring to the CEMAC zone’s common banking supervisor; four
 452 small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St.
 453 Vincent and the Grenadines, all 2025) are similarly recorded under “East-
 454 ern Caribbean States,” which we interpret as the Eastern Caribbean Central
 455 Bank’s member states. Both interpretations are flagged explicitly here as
 456 inferences rather than explicit per-country listings. Regulatory capacity is
 457 measured with the World Bank’s Worldwide Governance Indicators (WGI)
 458 Regulatory Quality estimate, matched to each economy’s exact survey year
 459 (or, where not yet published, the most recent prior year available) via the
 460 World Bank DataBank API; WGI does not cover Taiwan, which is accord-
 461 ingly dropped from global-sample regressions.

462 A natural alternative to the cross-sectional design used throughout this
 463 paper would exploit SBFN’s own staggered adoption timing—join years in
 464 our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences
 465 or event-study design around each economy’s own adoption date, the identi-
 466 fication strategy this literature (and this special issue specifically) most often
 467 asks for. We do not attempt this, and it is worth being explicit about why
 468 rather than leaving the gap implicit. The WBES Green Economy module
 469 is not a repeated panel instrument: it has been fielded to a given country,
 470 in most cases, exactly once, as a special one-off module attached to a single
 471 survey wave, and even the small subset of economies in our sample surveyed
 472 more than once by the WBES core instrument were not re-surveyed with a
 473 comparable Green Economy module in a second wave; the six supplementary

economies in our own extension sample (Section 5.5) illustrate the problem directly, since even they differ from one another, and from the primary sample, in exact item wording and randomized sub-module assignment, precluding a consistently defined outcome measured twice for the same country before and after its SBFN join date. A within-country before/after comparison around each economy’s own adoption date is accordingly not implementable with a consistent outcome variable in the currently published WBES microdata, for this specific outcome construct. This is a genuine data-availability constraint on the identification strategy available to us, not a design choice; we flag it here as a first-class limitation of the paper’s identification, and return to its consequences in Section 8.

4.2. Variable construction

Outcome. Our primary dependent variable, *green practice adoption*, is a binary indicator equal to one if a firm reports having adopted at least one of seven Green Economy module practices over the preceding three years: more climate-friendly on-site energy generation, an energy-management system, waste-minimization or recycling measures, air-pollution control measures, other pollution-control measures, any energy-efficiency measure, or the use of on-site renewable energy. We also construct a continuous *green adoption index*, the mean of the same seven binary items, for robustness.

Treatment. *Bank credit access* is a binary indicator equal to one if the firm reports holding a line of credit or loan from a formal financial institution at the time of the survey.

Country-level moderators. *SBFN member* is a binary indicator equal to one if the economy had adopted a national SBFN sustainable-finance

499 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
500 ulatory Quality estimate for the survey year (continuous, standardized to
501 mean zero, unit standard deviation within the estimation sample for the
502 hierarchical and machine-learning specifications).

503 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
504 own sampling strata), broad sector (manufacturing, services, construction),
505 log sales, an exporter dummy (any direct or indirect export share), and a
506 foreign-ownership dummy (any private foreign ownership share).

507 Table 1 lists every variable used in the analysis together with its exact
508 WBES question code and source, in the interest of full transparency about
509 variable construction given the multi-source nature of the merge.

510 4.3. Empirical strategy

511 We estimate the relationship between bank credit access and green prac-
512 tice adoption, and its moderation by SBFN status and regulatory quality,
513 in five stages of increasing methodological sophistication, each addressing a
514 specific limitation of the one before it.

515 **Stage 1 — classical benchmark.** We first estimate a pooled logistic
516 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

517 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a sec-
518 tor fixed effect, with standard errors clustered by country. We deliberately
519 do not include country fixed effects in this specification, and it is worth being
520 precise about what such a fixed effect would and would not absorb. Because

521 SBFN_c is constant within country in a single cross-sectional wave, a coun-
 522 try fixed effect absorbs its main effect entirely, making β_2 unidentifiable by
 523 construction. The interaction $\text{Credit}_{i,c} \times \text{SBFN}_c$, by contrast, is a firm-level
 524 regressor and remains formally identified under country fixed effects, since
 525 credit access itself varies within country; Stage 5 below relies on exactly that
 526 logic to identify the corresponding interactions on the global sample. What
 527 the primary sample cannot support is not the identification of β_3 but its pre-
 528 cision: with country fixed effects, β_3 would be identified only by the contrast
 529 between within-country credit variation in the eight SBFN-adopter economies
 530 of this sample and that in the remaining thirty-three (Section 5), while the
 531 SBFN main-effect level shift those economies exhibit—a first-order feature
 532 of the data—would be absorbed and unreportable. We therefore estimate
 533 β_2 and β_3 jointly without country fixed effects, as the classical benchmark
 534 against which Stage 2’s hierarchical alternative is compared, extended with
 535 a triple interaction for Regulatory quality_c that tests H3’s three-way claim
 536 directly.

537 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 538 estimate the same relationship allowing country to enter as a random rather
 539 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

540 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
 541 in PyMC, an open-source Python library for Bayesian statistical modelling
 542 (Salvatier et al., 2016). This specification is the paper’s primary vehicle for
 543 testing H2: because country enters through a variance component rather than
 544 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
 545 retain identified main effects, while the random slope $u_{1,c}$ still absorbs unob-
 546 served country heterogeneity in the credit-green relationship and, together
 547 with partial pooling across countries, gives more defensible inference than
 548 cluster-robust standard errors computed over only 41 clusters (Cameron and
 549 Miller, 2015). It carries two cross-level interactions: β_3 tests H2 directly, and
 550 β_4 tests whether regulatory quality on its own moderates the credit-green
 551 relationship. We are explicit that β_4 is not a test of H3. H3 asks whether
 552 the SBFN moderation in H2 is itself conditional on regulatory quality, which
 553 is a three-way claim requiring a $\text{Credit}_{i,c} \times \text{SBFN}_c \times \text{RegQuality}_c$ term; this
 554 specification does not include one, and we do not add it here because the pri-
 555 mary sample’s eight adopter economies cannot support a credibly estimated
 556 three-way interaction under partial pooling. H3 is instead tested directly
 557 by Stage 1’s fully saturated triple interaction above (Table 3, column M3)
 558 and, on an independent sample, by Stage 5’s no-fixed-effects specification
 559 (Section 5.7), both of which retain that term.

560 **Stage 3 — causal forest / double machine learning.** To avoid
 561 imposing a linear interaction functional form on how the treatment effect
 562 of credit access varies across the institutional landscape, we additionally es-
 563 timate a Causal Forest with Double Machine Learning (Wager and Athey,
 564 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted

565 nuisance models for the outcome and treatment propensity, and firm size, ex-
 566 port status, SBFN status, and regulatory quality as effect-modifying features.
 567 This lets us recover conditional average treatment effects (CATEs)—the esti-
 568 mated effect of credit access on green adoption, separately for SBFN members
 569 versus non-members, and separately by regulatory-quality tercile—without
 570 imposing that the moderation take the single-coefficient interaction form as-
 571 sumed in Stages 1–2. We use “causal” here in the qualified sense standard
 572 in this literature: the forest’s identifying assumption is unconfoundedness
 573 conditional on the observed covariate set (firm size, export status, SBFN
 574 status, regulatory quality), not an exogenous source of variation in credit
 575 access itself. This is exactly the same assumption underlying the logit and
 576 hierarchical specifications; Stage 3 relaxes the functional form imposed on
 577 that assumption from linear to fully flexible, but it does not relax the as-
 578 sumption, and the CATEs reported below should be read accordingly as
 579 doubly-robust conditional associations rather than causal effects identified
 580 off a natural experiment.

581 **Stage 4 — small-sample supplementary check.** We additionally
 582 re-estimate a reduced-form version of the credit-green relationship using a
 583 waste-minimization-adoption outcome, available in four economies (India
 584 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy
 585 modules asked this item under directly comparable wording. This outcome
 586 is not part of the standardized cross-country database underlying Stage 5
 587 below, so it offers genuinely independent, if small-sample, supplementary
 588 evidence.

589 **Stage 5 — global-sample replication with country fixed effects.**

590 Finally, and centrally, we re-estimate the full specification on the 162-economy
591 global sample described in Section 4, using CO2-emissions monitoring as the
592 outcome. Because this sample spans far more country clusters than the
593 primary sample, we can now afford a country-fixed-effects specification: δ_c
594 absorbs SBFN_c and RegQuality_c 's main effects (collinear with the fixed effect
595 by construction, since both are constant within country-year) while leaving
596 their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access
597 itself varies within country. This is a materially sharper test of H2 than Stage
598 1's clustered-SE approach, on an entirely independent sample and outcome
599 from Stages 1–3. The fixed-effects specification carries the two two-way in-
600 teractions only, so we report it alongside a no-fixed-effects specification for
601 two reasons: direct comparability with the primary sample's Table 3, and
602 because that specification, like Table 3's M3, retains the triple interaction
603 H3 requires.

604 5. Results

605 5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Bosnia and Herzegovina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

606 Table 2 reports the full 41-economy sample composition; firm counts range
607 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $> 0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $> 0\%$	D3B, D3C	WBES Sales Module

608 practice-adoption rate ranges more than three-fold across economies, from
609 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
610 SBFN policy status: 8 of the 41 primary-sample economies had adopted
611 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
612 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
613 economy sample once the five (of six) SBFN-member small-sample supple-
614 mentary economies are counted—and every European Union member state
615 in the sample is, consistent with SBFN’s mandate, a non-adopter. At the
616 firm level, bank credit access correlates positively with the green-adoption
617 index ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived
618 finance obstacle ($r = -0.02$)—a first hint, well before any regression, that re-
619 alized access to credit rather than perceived financing difficulty is what tracks
620 green behaviour, and the two are evidently not interchangeable measures of
621 the same underlying constraint.

622 Figure 2 previews the paper’s central finding graphically before a sin-
623 gle regression is estimated: plotting each economy’s green-adoption rate
624 against its regulatory-quality score and fitting separate lines for SBFN mem-
625 bers and non-members produces two lines of very similar, modestly positive
626 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
627 member economies show systematically *lower* raw green-adoption rates at
628 every level of regulatory quality, not a steeper relationship between regu-
629 latory quality and adoption. That combination—a level difference with no
630 slope difference—is exactly the signature the regression evidence below con-
631 firms formally.

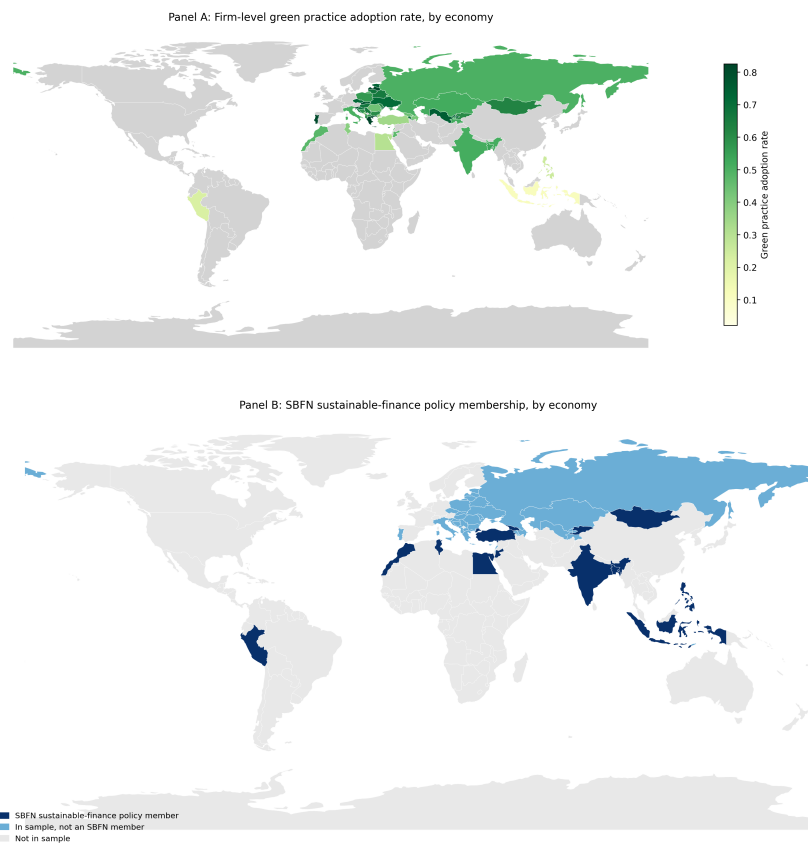


Figure 1: Green practice adoption and SBFN policy membership, by economy.

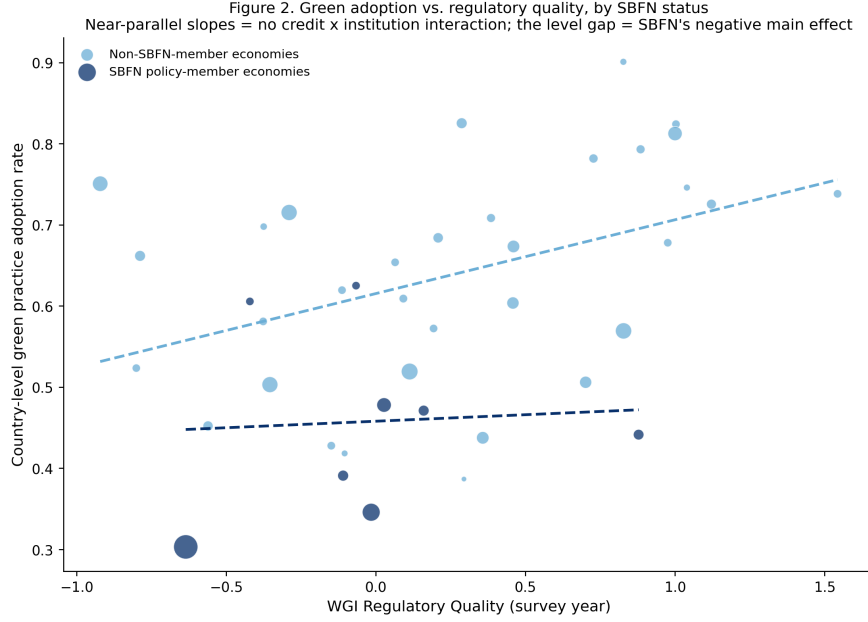


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3 adds the fully saturated triple interaction with regulatory quality, and

644 it is this column, not the hierarchical model of Section 5.3, that carries the
645 paper’s direct test of H3: the credit $\times$ SBFN $\times$ regulatory-quality term is
646 small, wrong-signed relative to H3’s prediction, and nowhere near significance
647 ($b = -0.104$, $se = 0.272$, $p = 0.702$). The rest of M3 tells the same story
648 at greater length: every interaction term involving credit access is small
649 and insignificant, while regulatory quality’s uninteracted relationship with
650 adoption is positive, if only marginally significant in this particular fully-
651 saturated specification ($b = 0.278$, $p = 0.110$).

652 As set out in Section 4, adding country fixed effects here would absorb the
653 SBFN and regulatory-quality main effects entirely, while leaving the interac-
654 tion terms identified but reliant on the contrast between within-country credit
655 variation in this sample’s eight adopter economies and its thirty-three non-
656 adopters. Table 3 accordingly reports main effects and interactions jointly,
657 without country fixed effects; what it shows is a genuine test of H2 and, in
658 M3, of H3, not a placeholder—and that test returns no interaction on ei-
659 ther. The question the remaining two stages are built to answer is whether
660 that null result reflects a real absence of policy moderation, or an artefact
661 of forcing a two-level question through a single-level model with country-
662 clustered standard errors computed over only 41 clusters, a setting in which
663 such standard errors are known to be unreliable (Cameron and Miller, 2015).

664 5.3. Bayesian hierarchical model — primary test of H2

665 The hierarchical specification—country-varying random slopes on credit
666 access, cross-level interactions with SBFN status and standardized regulatory
667 quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and
668 ess_{bulk} between 797 and 2,270—is built specifically to give H2 a fair hearing

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

under inference the 41-cluster classical specification cannot deliver, and to test regulatory quality’s own two-way moderation alongside it. H3’s three-way claim is not tested here (Section 4); it is tested in Table 3’s M3 above and in Section 5.7 below, and returns a null in both. This specification does not change the conclusion either. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN membership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one. The first of these is the direct H2 test; the second establishes that regulatory quality does not moderate the credit-green channel on its own, which is a necessary condition for the mechanism H3 posits without being a test of H3 itself.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we

694 can state that SBFN status and regulatory quality shift the *level* of green
695 adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

696 5.4. Causal forest / heterogeneous treatment effects

697 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
698 pose no linear-interaction functional form and let the treatment effect vary
699 flexibly across firm and country covariates. The overall average treatment
700 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
701 tent with, though smaller in magnitude and less precisely estimated than,
702 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more
703 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
704 ric estimator relative to a parametric logit on the same data. Critically,
705 the conditional average treatment effects triangulate with Section 5.3 rather
706 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
707 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /

0.075 across the low/middle/high regulatory-quality terciles respectively—
three different institutional cuts, three essentially flat effects, all visibly overlapping in Figure 5. Two entirely independent estimators, one parametric and hierarchical, one nonparametric and forest-based, agree on the absence of institutional moderation.

The forest’s feature-importance decomposition adds a genuinely new result rather than only confirming the null, and speaks directly to the auxiliary H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter status. Whatever heterogeneity exists in how strongly credit access predicts green adoption is, on this evidence, primarily a firm-level rather than a country-institutional phenomenon, supporting H4. Collapsing this into the Enterprise Surveys’ own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple “larger firms better absorb a green investment’s fixed costs” story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper’s institutional-moderation question,

733 and we return to it in Section 7.

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

734 *5.5. Small-sample supplementary check: waste-minimization adoption*

735 Before turning to the global-sample replication that is this paper’s second
736 major test (Section 5.7), Table 7 reports a smaller supplementary check using
737 waste-minimization adoption — an outcome not captured in the standardized
738 cross-country database underlying the global sample, and so a genuinely in-
739 dependent (if small-sample) data point — available in four economies whose
740 Green Economy modules asked this item under directly comparable word-
741 ing: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of
742 these four are themselves SBFN members at various progression stages, so

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

743 this sample cannot re-test the SBFN-adoption contrast; what it can test is
 744 whether the credit-to-green channel documented in Sections 5.2–5.4 is a pe-
 745 culiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes:
 746 pooled across the four economies, having an overdraft facility is associated
 747 with significantly higher odds of waste-minimization adoption ($b = 0.422$,
 748 $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green
 749 relationship replicates on a different continent, a different half-decade, and
 750 a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a
 751 preview, on four economies, of the much larger replication in Section 5.7.

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

752 5.6. Additional robustness

753 Table 9 subjects the primary-sample finding to four further checks, all
 754 retaining sector fixed effects, firm controls, and country-clustered standard

errors (no country fixed effects, for the reason given in Section 4). First, replacing the binary outcome with the continuous seven-item green-adoption index and estimating by OLS reproduces the same pattern to the decimal: credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistinguishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance measures produces one genuine point of nuance worth reporting rather than smoothing over. Using overdraft-facility access instead of a credit line/loan, the main overdraft effect remains positive and significant ($b = 0.274$, $p = 0.011$), the SBFN main effect remains negative and significant ($b = -0.905$, $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and significant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper that clears a conventional significance threshold, and it runs opposite to the direction H2 predicts. Using the reverse-coded perceived finance-obstacle measure in place of realized access, by contrast, both the main effect and its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interaction $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation noted in Section 5.1. We read the overdraft result as suggestive rather than conclusive given it is one significant coefficient among more than a dozen interaction tests across the paper, but it is at least directionally consistent with a substantive story worth flagging for future work: if SBFN-guided lending is channelled preferentially through the term loans and credit lines that finance long-horizon capital expenditure, rather than through short-term overdraft facilities, we might expect exactly this pattern—a null-to-positive

780 interaction on credit lines and a negative one on overdrafts—though our
781 data cannot directly test whether SBFN-guided credit is disproportionately
782 term-structured.

783 Third, splitting the sample into manufacturing ($N = 13,340$) and services
784 ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main
785 effect, and the null credit $\times$ SBFN interaction all replicate within each sector
786 separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b =$
787 -0.040 , $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b =$
788 0.195 , $p = 0.426$)—the paper’s central null is not an artefact of pooling two
789 structurally different production technologies.

790 Because the overdraft interaction (the one coefficient in the paper that
791 clears a conventional significance threshold) is one of many credit $\times$ institutional-
792 moderator tests reported across the frequentist specifications in this paper,
793 we additionally apply a Benjamini–Hochberg false-discovery-rate correction
794 across the full family of 13 such coefficients reported under a classical (non-
795 Bayesian, non-forest) estimator in Tables 3, 9, and 10 (Table 8; full calcu-
796 lation reported in the replication package). The overdraft interaction’s raw
797 $p = 0.031$ becomes $p = 0.403$ after correction, and no coefficient in the
798 family survives at a false discovery rate of 0.10 or even 0.40. We read this
799 as reinforcing, with a formal multiple-testing accounting rather than only a
800 qualitative caveat, the suggestive-not-conclusive framing we already give this
801 result in Section 7: an isolated nominally-significant interaction among more
802 than a dozen tested is exactly what a pure false-discovery process would be
803 expected to produce roughly one time in twenty, and this correction does not
804 let us reject that account.

Table 8: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (Tables 3, 9, 10). Table 3 model M3 entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table 9 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Table 10 M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Table 10 M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Table 3 M3 †	Credit $\times$ Reg. quality	0.310	0.799
Table 10 M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table 9 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Table 10 M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Table 3 M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table 9 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Table 3 M2	Credit $\times$ SBFN	0.773	0.892
Table 9 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table 9 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Table 3 M3 †	Credit $\times$ SBFN	0.892	0.892

Table 9: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

805 5.7. *Global-sample replication with country fixed effects*

806 The preceding five subsections establish the paper’s central finding on the
807 primary sample. We now subject that finding to what we consider the most
808 demanding test available to us: complete re-estimation on an independently
809 assembled, 162-economy global sample (Section 4), using a different outcome
810 variable (CO2-emissions monitoring rather than the seven-item composite),
811 different survey years (2021–2026 rather than 2018–2020), and, for the first
812 time in the paper, a country-fixed-effects specification that this much larger
813 set of clusters can support. Figure 3 maps the combined reach of the two
814 samples together: 166 distinct economies, spanning every region and, through
815 the global sample specifically, extending for the first time in this literature
816 into high-income economies with no SBFN involvement at all.

817 Table 10 reports two specifications. Column M1, without country fixed
818 effects, is the direct global-sample analogue of Table 3’s primary-sample base-
819 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,
820 $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
821 $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
822 every regulatory-quality interaction are statistically indistinguishable from
823 zero. Like Table 3’s M3, this specification is fully saturated and so retains
824 the credit $\times$ SBFN $\times$ regulatory-quality term H3 requires, which is likewise
825 insignificant; Table 10 tabulates the focal two-way terms only. H3 therefore
826 receives a direct, fully saturated test on both firm-level samples, and returns a
827 null on each. Column M2 imposes country fixed effects, absorbing the SBFN
828 and regulatory-quality main effects by construction (they are collinear with
829 δ_c , since both are constant within a given country-year) while leaving their

830 firm-level interactions with credit access fully identified, because credit access
831 itself varies within country. This is a sharper test than anything the primary
832 sample’s single 2018–2020 wave can support, and it returns the same answer
833 with even more precision: credit access is strongly significant ($b = 0.249$,
834 $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
835 the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
836 null.

837 We regard this as the paper’s most important robustness result. It is not
838 a re-analysis of the same data under a different label: it is a different outcome
839 construct, a different set of economies (only a handful of which overlap with
840 the primary sample), a different half-decade, and an econometric specification
841 — country fixed effects — that is only available at all because this second
842 sample has enough clusters to support it. That the qualitative conclusion
843 is identical across both samples and across the full set of four estimators
844 reported in this paper (classical logit, Bayesian hierarchical, causal forest, and
845 now country-fixed-effects logit) is, we think, considerably stronger evidence
846 for the paper’s central claim than either sample could offer alone.

847 *5.8. Results summary across all specifications*

848 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
849 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
850 into a single view. Panel A shows the credit-access effect (H1): every point
851 estimate is positive and every confidence interval excludes zero, in both sam-
852 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
853 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
854 terval, across both samples and every estimator, straddles zero. Figure 5

Table 10: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

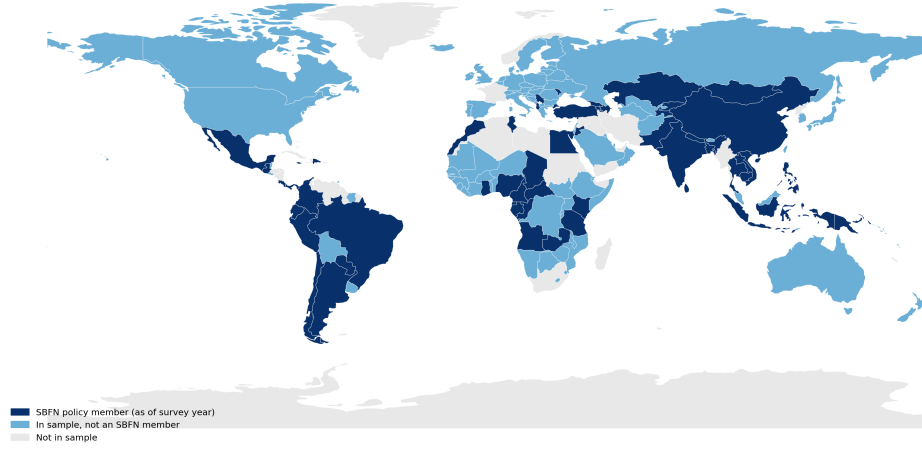


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

855 shows the same non-result from a completely different angle—the causal
 856 forest’s conditional average treatment effects broken out by SBFN status,
 857 regulatory-quality tercile, and firm-size category all overlap both each other
 858 and the overall average treatment effect. No subgroup split, by any variable
 859 examined in this paper, recovers a distinguishable effect.

Figure 4. Coefficient estimates across every specification and both samples
 (Panel A: robust and significant everywhere; Panels B-C: null everywhere)

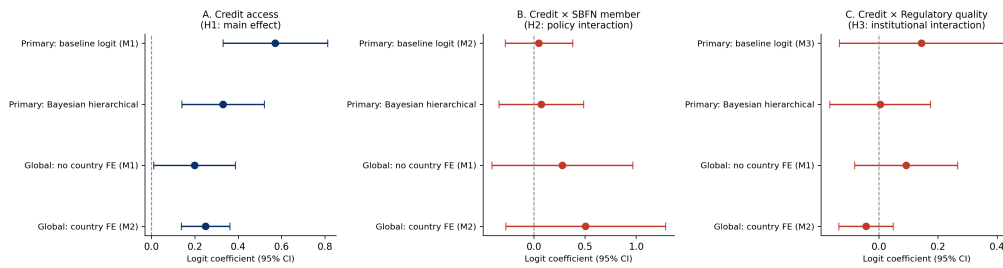


Figure 4: Coefficient estimates across every specification and both samples.

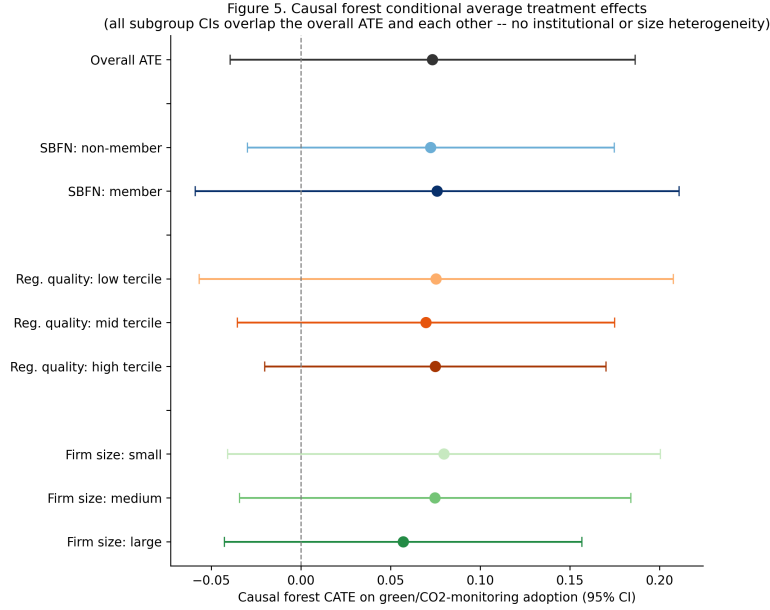


Figure 5: Causal forest conditional average treatment effects, by subgroup.

860 6. Supplementary evidence: a country-level staggered event-study 861 on macro environmental outcomes

862 Sections 5.2–5.8 establish this paper’s central finding using firm-level
863 cross-sectional data, for the data-availability reasons given in Section 4: the
864 WBES Green Economy module cannot support a staggered difference-in-
865 differences or event-study design around each economy’s own SBFN adop-
866 tion date, because it has not been fielded as a repeated panel instrument.
867 That constraint applies to the firm-level finance-to-green-investment channel
868 specifically; it does not apply to a different, complementary question that
869 the same SBFN adoption-timing variation can answer directly: does a coun-
870 try’s aggregate environmental trajectory shift around its own SBFN adoption
871 date? This section builds and reports that test, using an entirely indepen-

dent data source and unit of analysis, precisely because it is the identification strategy this literature – and this special issue in particular – asks for, and because a genuine answer, however it comes out, is more useful than leaving the question unaddressed.

6.1. Data and design

We assemble a country-year panel spanning every World Bank member economy with a non-aggregate ISO3 code (217 economies), 2000–2024. Treatment timing is each economy’s SBFN join year, drawn from the same SBFN data portal membership roster used to construct the global sample (Section 4) and cross-checked against the global-sample SBFN coding used in Section 5.7 (zero disagreements across 61 overlapping economies); economies that have never joined SBFN, as of the roster’s most recent update, serve as the never-treated comparison group. This yields 68 ever-treated economies (join years 2012–2024, the same real staggered-adoption variation described in Section 2) and 149 never-treated economies. Outcome data come from the World Bank’s World Development Indicators, retrieved via the World Bank DataBank API: renewable energy consumption (% of total final energy consumption, `EG.FEC.RNEW.ZS`) and CO2 emissions per capita (AR5 GWP-consistent series, `EN.GHG.CO2.PC.CE.AR5`), both available for the large majority of economies from 2000 through at least 2021–2024. We additionally draw on GDP per capita (constant 2015 US\$, `NY.GDP.PCAP.KD`) as a covariate, to net out the obvious confound that SBFN-adopting economies are disproportionately still-industrializing, faster-growing economies whose emissions profiles are shaped by their development stage independently of any banking-sector policy.

897 We estimate the effect of SBFN adoption on each outcome using the
 898 Callaway and Sant’Anna (2021) staggered-adoption difference-in-differences
 899 estimator (Callaway and Sant’Anna, 2021), which recovers group-time av-
 900 erage treatment effects for each adoption cohort separately and aggregates
 901 them into an overall average treatment effect on the treated (ATT) and
 902 an event-study profile. We report this estimator, rather than a standard
 903 two-way-fixed-effects (TWFE) regression, because TWFE is now well known
 904 to produce badly biased estimates under staggered treatment timing when
 905 already-treated units are implicitly used as controls for later-treated units
 906 (Goodman-Bacon, 2021); we report a naive TWFE specification alongside
 907 the Callaway-Sant’Anna estimates specifically to illustrate the size of that
 908 bias in our own data, rather than as a candidate preferred estimate. For
 909 each outcome we report both an unconditional specification and a doubly-
 910 robust specification conditioning on log GDP per capita, and we check the
 911 doubly-robust specification’s sensitivity to the choice of comparison group,
 912 using never-treated economies as our preferred comparison group throughout
 913 and not-yet-treated economies (those that will eventually adopt but had not
 914 yet, as of the relevant relative period) as a robustness check.

915 *6.2. Results*

916 Table 11 reports the overall ATT for both outcomes across four specifica-
 917 tions, including a comparison-group robustness check; Figure 6 plots the full
 918 event-study profile (doubly-robust, GDP-per-capita-controlled, never-treated
 919 specification) for both outcomes.

920 Two results, read together, do not support treating SBFN adoption as a
 921 country-level lever that visibly improves aggregate environmental outcomes –

Table 11: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no covariates, never-treated	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.653	0.919	[-1.147, 2.454]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no covariates, never-treated	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.313**	0.148	[0.023, 0.603]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.

Country-level staggered event-study, SBFN adoption on macro environmental outcomes
(Callaway-Sant'Anna, doubly-robust, log GDP per capita control; shaded = pointwise 95% CI;
relative periods beyond $\pm 12/9$ trimmed, too few surviving cohorts to be informative)

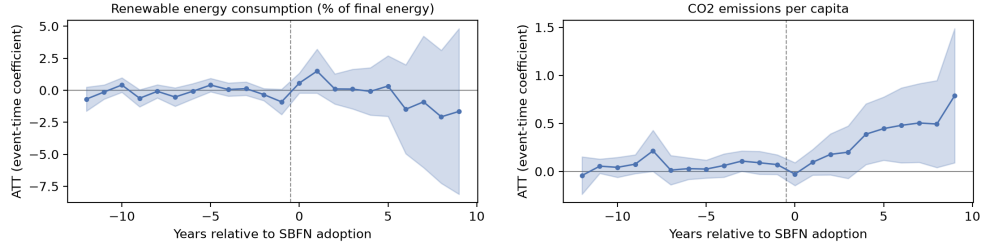


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

922 which is itself informative, since this is the identification strategy best suited
 923 to answer that specific question directly. First, renewable energy share shows
 924 no effect distinguishable from zero in either specification, and the doubly-
 925 robust event-study profile (Figure 6, left panel) is flat and precisely estimated
 926 on both sides of the adoption date: a clean null, not an underpowered one.

927 Second, CO2 emissions per capita shows a positive (i.e., the “wrong-
 928 signed” direction if SBFN adoption reduced emissions) and statistically sig-
 929 nificant association with adoption in every specification, but three features
 930 of the result argue against reading it causally. It shrinks by roughly 45%
 931 (0.631 to 0.349) once we condition on log GDP per capita, indicating that
 932 a substantial share of the raw association is attributable to exactly the
 933 development-stage confound we set out to net out. The doubly-robust event-
 934 study profile (Figure 6, right panel) shows a small, mostly flat pre-period
 935 with only two marginally significant leads, followed by a post-period effect
 936 that grows smoothly and monotonically from near zero at adoption to its

937 largest magnitude nine to twelve years out – a continuing-divergence pattern
938 more consistent with SBFN-adopting economies already sitting on a different,
939 faster-rising emissions trajectory than non-adopters (which a single covariate
940 cannot fully absorb) than with a discrete policy-caused jump at the adop-
941 tion date. And the naive TWFE estimate (0.901) is itself more than double
942 the doubly-robust Callaway-Sant’Anna estimate (0.349), the same direction
943 of bias documented in the staggered-DID literature (Goodman-Bacon, 2021)
944 and a useful internal illustration of why we lead with the more robust esti-
945 mator throughout this section.

946 Both doubly-robust results are stable to the choice of comparison group:
947 substituting not-yet-treated economies for never-treated economies leaves the
948 CO2 estimate essentially unchanged (0.313 versus 0.349, both significant at
949 conventional levels) and the renewable-energy null equally unchanged (0.653
950 versus 0.674, neither distinguishable from zero); Table 11 reports both. Nei-
951 ther conclusion in this section depends on which set of untreated economies
952 serves as the counterfactual.

953 Neither result supports a “green banking policy backfires” reading, and
954 neither supports a “green banking policy works” one. The more measured
955 reading is that a macro-level design – built using exactly the identification
956 strategy (staggered adoption timing, a proper Callaway-Sant’Anna estima-
957 tor, never-treated controls) this special issue’s own methodological expecta-
958 tions call for – also does not produce clean evidence that SBFN adoption
959 changes a country’s aggregate environmental trajectory for the better, and
960 the one statistically significant result we do find is more parsimoniously ex-
961 plained by which economies self-select into SBFN membership than by what

the policy itself does once adopted. This is, we think, a genuinely different and complementary piece of evidence to Sections 5.2–5.7’s firm-level results, not a restatement of them: different data source, different unit of analysis, different outcome constructs, and, for the first time in this paper, an identification strategy that exploits exogenous variation in adoption *timing* rather than relying on cross-sectional comparison across adopters and non-adopters. That it converges on the same substantive conclusion – no clean evidence of a beneficial institutional effect – from a design built specifically to meet a bar the paper’s firm-level evidence cannot meet, strengthens rather than substitutes for the paper’s central finding.

7. Discussion

Put in plain economic terms, the paper’s central finding is this: a firm holding a bank credit line is associated with an adoption probability roughly 13 percentage points higher than an otherwise-similar firm without one—an association net of observed controls, not a causally identified effect (Limitation 1)—and that gap is essentially unchanged whether the firm sits in an SBFN adopter or non-adopter, or across the full range of regulatory quality (Figure 4). Firms in SBFN-adopting economies are, on average, 15–20 percentage points *less* likely to have adopted a green practice at all—a level gap, not a slope gap.

This convergence is informative precisely because of how differently the evidence could have failed to agree. The primary and global samples share almost no economies, span different half-decades, use different outcome items, and reach institutional settings—high-income, non-SBFN economies such as

986 Australia and Germany—the primary sample cannot touch. Section 6’s
987 country-level staggered event-study adds a third, independent test on top
988 of this: a different data source, a different unit of analysis, and the paper’s
989 only design that exploits exogenous variation in adoption *timing*. That it
990 too finds no clean evidence of a beneficial institutional effect—a precise null
991 on renewable-energy share, and a CO2 association that shrinks substantially
992 once development stage is controlled for—is the hardest-to-dismiss evidence
993 that this is a real feature of the data, not an artefact of any one sample, level
994 of analysis, or estimator. (This is a distinct claim from H2/H3’s firm-level
995 moderation null—a main effect at country level, not a slope change at firm
996 level—but evidence of the same kind.)

997 This complicates rather than confirms the implicit premise of a green-
998 credit-policy literature disproportionately built on China’s unusually mature,
999 well-resourced guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025).
1000 Our sample spans every stage of SBFN’s own progression framework, from
1001 Jordan and the Kyrgyz Republic’s initial “Commitment” to Mongolia’s “Ad-
1002 vancing,” and across that range we find no evidence that formal adoption
1003 changes how credit access translates into green investment. The causal for-
1004 est’s feature-importance decomposition (Section 5.4) offers one redirection
1005 rather than a tidy answer: the heterogeneity that does exist is overwhelm-
1006 ingly a firm-size phenomenon (75.6% of explained variance) rather than in-
1007 stitutional (regulatory quality 19.9%, SBFN 2.6%)—though it does not re-
1008 solve into a clean size gradient, so we can say where the heterogeneity sits
1009 without yet saying in which direction it runs. If the binding constraint is
1010 which firms can absorb a green investment’s fixed costs once financed, rather

1011 than whether a supervisory guideline nudges banks to lend more broadly, a
1012 country-level mandate on the extensive margin of lending may be aimed at
1013 the wrong margin regardless of its own credibility.

1014 “No interaction detected” is not the same claim as “no institutional com-
1015 plementarity exists,” and we do not want the former to be read as settling
1016 the latter. Two alternative readings stay live: a linear interaction may be
1017 the wrong functional form for a genuinely threshold-shaped complementarity,
1018 given SBFN’s own graded progression framework and our binary coding of
1019 it (Limitation 2); and the Worldwide Governance Indicators may simply not
1020 proxy the banking-supervisory capacity a green-lending guideline depends
1021 on (Limitation 5). We think our preferred reading—no complementarity
1022 through this channel—is best supported by the evidence, but hold it as the
1023 best current account rather than a claim that rules the alternatives out.

1024 Two further limits are worth naming directly. First, SBFN guidelines
1025 could in principle be working on the *extensive* margin of who obtains credit
1026 at all, rather than the *intensive* margin our interaction terms test; testing
1027 that would require modelling credit access itself as a function of SBFN status,
1028 which our demand-side design does not do. Second, every outcome in this
1029 paper is firm-reported, and we cannot rule out a compliance-reporting-versus-
1030 practice gap between what firms report and what banks report to their own
1031 supervisor—a limit no firm-level survey instrument can fully resolve on its
1032 own. Read together with Section 6, whose null is, to our knowledge, the first
1033 test of the SBFN framework’s own theory of change against outcomes on
1034 the ground rather than its own tracked policy-stage progression, both point
1035 the same way: a framework can look like it is working by every measure the

1036 network itself tracks while the channel it is meant to move remains, on this
1037 evidence, unmoved.

1038 The one interaction that clears a conventional significance threshold—a
1039 negative credit $\times$ SBFN effect for overdraft rather than term-credit access
1040 (Section 5)—is more consistent with this redirection than with a simple policy
1041 failure: overdrafts finance working capital, not the long-horizon expenditure
1042 our outcome measures, so a supervisor successfully steering credit toward
1043 term instruments would produce exactly this pattern. We flag it as suggestive
1044 only: it is one significant coefficient among many tested, and does not survive
1045 a Benjamini–Hochberg correction at any conventional false discovery rate
1046 (Table 8).

1047 Read against this special issue’s themes of state capacity and policy cred-
1048 ibility, our results sit closer to a state-capacity-is-necessary-but-firm-level-
1049 frictions-bind-independently story than a straightforward complementarity
1050 one. Regulatory quality raises the baseline propensity to adopt green prac-
1051 tices, but is not the missing ingredient that lets a green banking guideline
1052 reshape the credit-to-green channel specifically. The binding constraint looks
1053 less like whether the supervising jurisdiction can make its guideline credible,
1054 and more like whether firms below a certain scale can absorb a green invest-
1055 ment’s fixed costs once credit is available—a question one level down from
1056 the one the SBFN framework is designed to answer.

1057 8. Conclusion

1058 This paper set out to test whether the wave of green banking policy
1059 adoption coordinated through the Sustainable Banking and Finance Net-

1060 work changes the relationship between firm-level access to finance and green
1061 investment behaviour, and whether that depends on the regulatory capacity
1062 available to make the policy credible. Using harmonized WBES data across
1063 41 economies (2018–2020), a four-country waste-minimization check, and an
1064 independently assembled 162-economy global sample (2021–2026) reaching
1065 for the first time into high-income, non-SBFN economies, we apply four es-
1066 calating estimators and find five consistent results. First, bank credit access
1067 is robustly associated with green practice adoption, surviving every specifi-
1068 cation, subsample, and alternative-outcome check, and replicating in both
1069 the small waste-minimization sample and the full global sample. Second,
1070 neither SBFN adoption nor regulatory quality moderates that relationship
1071 under any estimator, on either sample, though both shift the general level of
1072 adoption directly in the primary sample. Third, where effect heterogeneity
1073 does exist, it sits at the level of firm size rather than country institutions.
1074 Fourth, this pattern replicates on a sample sharing almost no economies with
1075 the primary one, under a materially sharper identification strategy. Fifth,
1076 and most importantly for the paper’s credibility, an independent country-
1077 level staggered event-study (Section 6)—a main-effect test, not a repeat of
1078 the firm-level moderation null—likewise finds no clean evidence that SBFN
1079 adoption improves aggregate environmental outcomes, the single strongest
1080 evidence that this null is a real feature of the data rather than an artefact of
1081 any one sample, data source, level of analysis, or estimator.

1082 For policy, the implication is not that green banking guidelines are in-
1083 effective, but that our evidence does not support treating them as a lever
1084 for widening the green-investment gap between financed and unfinanced

1085 firms—the margin most of this literature implicitly assumes they operate
1086 on. If our tentative loan-tenor reading of the overdraft result is correct,
1087 policymakers may find more traction focusing on the term-lending instru-
1088 ments that actually finance green capital expenditure, and on complementary
1089 measures—technical assistance, matching grants, credit guarantees—that ad-
1090 dress smaller firms’ absorptive-capacity constraints directly.

1091 We close with seven limitations. First, both firm-level samples are single
1092 cross-sections (Section 4 explains why a genuine panel does not exist for this
1093 outcome): they cannot rule out reverse causality or unobserved country char-
1094 acteristics correlated with both SBFN adoption and baseline green-adoption
1095 propensity, and the global sample’s fixed effects sharpen this only for the
1096 interaction terms, not the credit main effect. Second, SBFN membership
1097 is a binary proxy for what the network’s own framework treats as a graded
1098 process; the global sample’s regional roster entries (the CEMAC and East-
1099 ern Caribbean groupings, Section 4) additionally rest on an inference about
1100 coverage rather than an explicit per-country statement. Third, the four-
1101 economy waste-minimization check cannot re-test the SBFN contrast (three
1102 of its four economies are members) and combines non-identical outcome items
1103 by construction of WBES’s own shortened module design—a data-availability
1104 constraint, not a methodological choice. Fourth, the global sample’s CO2-
1105 monitoring outcome is a single item, narrower than the primary sample’s
1106 seven-item composite; that the same pattern holds on it is reassuring but
1107 does not establish it would hold on a richer measure. Fifth, the World-
1108 wide Governance Indicators measure is a broad governance indicator, not a
1109 banking-supervision-specific capacity measure. Sixth, our design tests only

1110 the intensive margin of the credit-to-green channel and cannot rule out that
1111 SBFN policy operates on the extensive margin of who obtains credit, nor dis-
1112 tinguish firm-reported practice adoption from a bank-compliance-reporting
1113 gap (Section 7). Seventh, Section 6’s event-study is an ecological-level test
1114 of a related but distinct question; its one significant result (CO2 per capita)
1115 is netted against a single covariate, and we cannot rule out that a richer set
1116 of development-stage controls would shrink it further, or that some genuine,
1117 small, confounded relationship remains. We leave each of these to future
1118 work.

1119 **Declaration of generative AI and AI-assisted technologies in the** 1120 **manuscript preparation process**

1121 During the preparation of this work the author(s) used AI-assisted tools
1122 to help polish the writing. After using these tools, the author(s) reviewed
1123 and edited the content as needed and take full responsibility for the content
1124 of the published article.

1125 **Appendix A. SBFN policy status and regulatory quality, full sam-** 1126 **ple**

1127 Table A.12 reports, for all 47 economy-years in the paper’s primary and
1128 small-sample supplementary check, the SBFN membership status coded for
1129 the analysis, the join year where applicable, the SBFN Multi-Tiered Progress
1130 Framework stage in force nearest the survey year, and the exact year of the
1131 Worldwide Governance Indicators observation used. Full per-country source
1132 citations (the specific SBFN Global Progress Report, country addendum, or

1133 accession announcement consulted for every determination) are available in
1134 the online supplementary material. The analogous 162-economy table for
1135 the global sample (Section 4) is summarized by region in Appendix C and
1136 provided in full in the online supplementary material and the replication
1137 package.

Table A.12: SBFN policy status and regulatory quality, full 47-
economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integra- tion)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Prepara- tion)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementa- tion)	2019
Greece 2018	0	—	None	2018

Economy	SBFN member	Join year	Stage at survey		WGI year
Hungary 2019	0	—	None		2019
India 2022	1	2016	Developing	(Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)		2023
Italy 2019	0	—	None		2019
Jordan 2019	1	2016	Commitment	(Preparation)	2019
Kazakhstan 2019	0	—	None		2019
Kosovo 2019	0	—	None		2019
Kyrgyz Republic 2019	1	2018	Commitment	(Preparation)	2019
Latvia 2019	0	—	None		2019
Lebanon 2019	0	—	None		2019
Lithuania 2019	0	—	None		2019
Malta 2019	0	—	None		2019
Moldova 2019	0	—	None		2019
Mongolia 2019	1	2012	Advancing	(Implementation)	2019
Montenegro 2019	0	—	None		2019
Morocco 2019	1	2014	Advancing	(Implementation)	2019
North Macedonia 2019	0	—	None		2019
Peru 2023	1	2013	Advancing	(ESG Integration)	2023
Philippines 2023	1	2013	Advancing	(ESG Integration)	2023

Economy	SBFN member	Join year	Stage at survey	WGI year
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Republic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementation)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

1138 Appendix B. Bayesian model convergence diagnostics

1139 Convergence diagnostics for every focal parameter of the hierarchical
1140 model reported in Table 4 are included directly in that table ($\hat{R}$ column); the
1141 full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member)
1142 to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional
1143 minimum threshold of 400 effective samples per parameter for stable poste-

rior summaries at four chains. No divergent transitions were flagged during sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup draws each after 1,000 tuning iterations).

Appendix C. Global-sample SBFN status, regional summary

Of the 162 global-sample economy-years, 61 are coded as SBFN members as of their survey year and 101 as non-members. Table C.13 reports the regional distribution of the 61 members, drawn from the SBFN data portal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste, Bangladesh, and Indonesia) carry the deeper per-country sourcing already reported in Appendix A, having been determined during the primary sample’s original research pass rather than from the portal roster alone.

Table C.13: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

1155 **Additional material**

1156 *Online appendix*

1157 Appendix A, Appendix B, and Appendix C appear above; per-country
1158 source citations at the level of detail compiled during data construction (in-
1159 dividual SBFN Global Progress Report page references, country-addendum
1160 URLs, and IFC accession-announcement links for every non-obvious deter-
1161 mination in the primary sample; the full 162-row global-sample SBFN/WGI
1162 table underlying Appendix C) are available in the online supplementary
1163 material accompanying this submission.

1164 *Data availability*

1165 Data in support of the findings of this study is available from the corre-
1166 sponding author upon reasonable request, subject to the World Bank Enter-
1167 prise Surveys' own terms of use for the underlying firm-level microdata (enter-
1168 prisesurveys.org, free registration required), including the standardized cross-
1169 country database and Productivity Database underlying the global sample.
1170 The country-level SBFN policy-status (including the SBFN data portal's
1171 member roster underlying the global sample) and Worldwide Governance In-
1172 dicators datasets constructed for this paper, the country-year World Develop-
1173 ment Indicators panel and Callaway-Sant'Anna event-study code underlying
1174 Section 6 (all freely redistributable, no registration required), together with
1175 the full analysis and replication code, are available in a public GitHub repos-
1176 itory; the repository link is withheld from this anonymized manuscript, and
1177 repository access is restricted for the duration of double-anonymized review,
1178 to avoid identifying the authors to reviewers. The link will be provided to

1179 the editor upon submission and made publicly accessible again to readers
1180 upon acceptance.

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